

HW4 (Chapters 6) for MSE 201-002 Spring 2023

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8th Edition

Chapter 6

Problem 6.8

A cylindrical rod of copper ($E = 110 \text{ GPa}$, $16 \times 10^6 \text{ psi}$) having a yield strength of 240 MPa ($35,000 \text{ psi}$) is to be subjected to a load of 6660 N (1500 lb_f). If the length of the rod is 380 mm (15.0 in.), what must be the diameter to allow an elongation of 0.50 mm (0.020 in.)?

Problem 6.22

Consider the brass alloy for which the stress–strain behavior is shown in Figure 6.12. A cylindrical specimen of this material 6 mm (0.24 in.) in diameter and 50 mm (2 in.) long is pulled in tension with a force of 5000 N (1125 lb_f). If it

is known that this alloy has a Poisson's ratio of 0.30 , compute **(a)** the specimen elongation and **(b)** the reduction in specimen diameter.

Problem 6.29

A cylindrical specimen of aluminum having a diameter of 0.505 in. (12.8 mm) and a gauge length of 2.000 in. (50.800 mm) is pulled in tension. Use the load–elongation characteristics shown in the following table to complete parts (a) through (f).

<i>Load</i>		<i>Length</i>	
<i>N</i>	<i>lb_f</i>	<i>mm</i>	<i>in.</i>
0	0	50.800	2.000
7,330	1,650	50.851	2.002
15,100	3,400	50.902	2.004
23,100	5,200	50.952	2.006
30,400	6,850	51.003	2.008
34,400	7,750	51.054	2.010
38,400	8,650	51.308	2.020
41,300	9,300	51.816	2.040
44,800	10,100	52.832	2.080
46,200	10,400	53.848	2.120
47,300	10,650	54.864	2.160
47,500	10,700	55.880	2.200
46,100	10,400	56.896	2.240
44,800	10,100	57.658	2.270
42,600	9,600	58.420	2.300
36,400	8,200	59.182	2.330
Fracture			

- (a) Plot the data as engineering stress versus engineering strain.
- (b) Compute the modulus of elasticity.
- (c) Determine the yield strength at a strain offset of 0.002.
- (d) Determine the tensile strength of this alloy.
- (e) What is the approximate ductility, in percent elongation?
- (f) Compute the modulus of resilience.

Problem 6.41

Using the data in Problem 6.29 and Equations 6.15, 6.16, and 6.18a, generate a true stress–true strain plot for aluminum. Equation 6.18a becomes invalid past the point at which necking begins; therefore, measured diameters are given in the following table for

the last four data points, which should be used in true stress computations.

<i>Load</i>		<i>Length</i>		<i>Diameter</i>	
<i>N</i>	<i>lb_f</i>	<i>mm</i>	<i>in.</i>	<i>mm</i>	<i>in.</i>
46,100	10,400	56.896	2.240	11.71	0.461
44,800	10,100	57.658	2.270	11.26	0.443
42,600	9,600	58.420	2.300	10.62	0.418
36,400	8,200	59.182	2.330	9.40	0.370

Problem 6.D2

6.D1 A large tower is to be supported by a series of steel wires. It is estimated that the load on each wire will be 11,100 N (2500 lb_f). Determine the minimum required wire diameter, assuming a factor of safety of 2 and a yield strength of 1030 MPa (150,000 psi).

6.D2 (a) Gaseous hydrogen at a constant pressure of 1.013 MPa (10 atm) is to flow within the inside of a thin-walled cylindrical tube of nickel that has a radius of 0.1 m. The temperature of the tube is to be 300°C and the pressure of hydrogen outside of the tube will be maintained at 0.01013 MPa (0.1 atm). Calculate the minimum wall thickness if the diffusion flux is to be no greater than 1×10^{-7}

mol/m²·s. The concentration of hydrogen in the nickel, C_H (in moles hydrogen per m³ of Ni), is a function of hydrogen pressure, P_{H_2} (in MPa), and absolute temperature (T) according to

$$C_H = 30.8 \sqrt{P_{H_2}} \exp \left(-\frac{12.3 \text{ kJ/mol}}{RT} \right) \quad (6.28)$$

Furthermore, the diffusion coefficient for the diffusion of H in Ni depends on temperature as

$$D_H(\text{m}^2/\text{s}) = 4.76 \times 10^{-7} \exp \left(-\frac{39.56 \text{ kJ/mol}}{RT} \right) \quad (6.29)$$

(b) For thin-walled cylindrical tubes that are pressurized, the circumferential stress is a function of the pressure difference across the wall (Δp), cylinder radius (r), and tube thickness (Δx) as

$$\sigma = \frac{r \Delta p}{4 \Delta x} \quad (6.30)$$

Compute the circumferential stress to which the walls of this pressurized cylinder are exposed.

(c) The room-temperature yield strength of Ni is 100 MPa (15,000 psi) and, furthermore, σ_y diminishes about 5 MPa for every 50°C rise in temperature. Would you expect the wall thickness computed in part (b) to be suitable for this Ni cylinder at 300°C? Why or why not?

(d) If this thickness is found to be suitable, compute the minimum thickness that could be

used without any deformation of the tube walls. How much would the diffusion flux increase with this reduction in thickness? On the other hand, if the thickness determined in part (c) is found to be unsuitable, then specify a minimum thickness that you would use. In this case, how much of a diminishment in diffusion flux would result?

6.D3 Consider the steady-state diffusion of hydrogen through the walls of a cylindrical nickel tube as described in Problem 6.D2. One design calls for a diffusion flux of 5×10^{-8} mol/m²·s, a tube radius of 0.125 m, and inside and outside pressures of 2.026 MPa (20 atm) and 0.0203 MPa (0.2 atm), respectively; the maximum allowable temperature is 450°C. Specify a suitable temperature and wall thickness to give this diffusion flux and yet ensure that the tube walls will not experience any permanent deformation.